

RESEARCH INTERESTS

Industrial Automation and Digital Manufacturing

- Machine Vision and Generative AI for predicting product quality through automated visual inspections
- Cloud-Edge / IoT solutions for process automation
- Operational and Human digital twins for safe human-machine interactions in the physical world
- Industrial Augmented / Virtual / Extended Reality for digital work instructions to aid worker training
- Synthetic data generation for deep learning applications

EDUCATION

Ph.D., University of Cincinnati, OH, USA. GPA 4.0 (Advisor: Prof. Sam Anand) 2018 – Present
Dissertation: Machine Learning and IIoT-assisted Computer Vision Frameworks for Smart and Safe Manufacturing

M.S., Mechanical Engineering, Michigan Technological University, MI, USA. 2015 – 2017

B.S., Mechanical Engineering, University of Pune, India. 2010 – 2014

INDUSTRY PROJECTS

Siemens Technology, NJ, USA

[Modeling Human Motion for Improving Worker Safety in Manufacturing Environments](#) 2022 - Present

- Trained a multi-task GPT model for automatically:
 - generating a 3D animation illustrating how to perform a new task ergonomically,
 - providing textual feedback on whether the worker's current motion could lead to injury,
 - predicting the worker's projected path to prevent any object collisions
- Devised LLM-assisted toolbox to label custom 3D human motions of factory workers with ergonomic descriptions
- Project group awarded \$1.3M by Ohio Bureau of Workers' Compensation to deploy at selective industrial sites

[Automated Quality Control using AI and Computer Vision \(CV\)](#)

- Developed a closed-loop CV system for automated, real-time inspection of sealant application for space-grade solar panel
- Performed object localization and image segmentation of the sealant deposited on plates, and trained a Bayesian network to provide corrective action recommendations.
- Deployed at Boeing facility, Charleston, SC as part of DMDII/MxD funding (\$1.5M)

Siemens, P&G, Kinetic Vision, Worthington Steel (Selective Industry 4.0/5.0 Institute members at UC) 2023 - 2024

[Enhancing Digital Twins and IoT for Legacy and Smart Factory Machines with Large Language Model \(LLM\) Feedback](#)

- Co-developed LLM-based digital twin for guiding new users to control factory machines (on-site / remotely) through voice commands
- Proposed a RAG-based multi-agent framework to process IoT data for visualizing plots, generating documentation for machine operations, conducting fault diagnosis, and suggesting potential fixes (Latency < 1.5 seconds)

[Improving Worker Manufacturing Training with Augmented Reality and Deep Learning](#)

- Developed an AR-based training framework that simplifies complex component identification for shop floor workers
- Built a novel synthetic data generation pipeline, based on CAD models using physics-based rendering methods, and fine-tuned a deep learning model (U-net) for pixel-level component identification and performing assembly

Volvo Trucks 2021

[Real-Time Visualization of Machine Sensors for Factory Maintenance using Cloud and AR Application](#)

- Co-developed an AR methodology to locate sensor positions (spatial anchors) on machines and visualize data in real-time to aid workers
- Reduced machine troubleshooting and downtime by 30%, translating to several million dollars per year in savings

Raytheon Technologies 2020

[CAD-Based Computational Tools for Metal Powder Additive Manufacturing](#)

- Developed computational geometry algorithms for feature recognition in 3D geometries
- Designed frontend application GUI within Siemens NX

Tata Motors Ltd. 2015

[Engineering Intern, Research and Development](#)

- Designed rapid prototype and sheet metal components in CAD for 3D printing and press bending operations

RELEVANT RESEARCH PROJECTS

Industrial IoT for Smart Factory

- Created an edge and cloud-based AR framework to display and control real-time legacy machine data on an iPad and provide corrective actions using a Bayesian Network; awarded \$40K grant from UC Industry 4.0/5.0 Institute to build a Digital Twin

Prediction of Selective Laser Melting (3D Printing) Part Quality using Hybrid Bayesian Network

- Devised a Bayesian network to model the correlation between process parameters and part quality attributes of SLM machines

Deep Learning Segmentation Methods for Defect Detection in Additive Manufacturing

- Published a survey at Int. Journal of Advanced Manufacturing Technology on CV methods reported for 3D printing using YOLO, SSD, Faster R-CNN, U-Net, Mask R-CNN, GAN, and Vision Transformers and implementing them for closed-loop monitoring

PROGRAMMING AND SOFTWARE SKILLS

Programming languages: Python, C#, C++, MATLAB, JavaScript, HTML, SQLite
Commercial software: NX, Siemens Jack/Process Simulate, Ansys Additive, Autodesk Netfabb, Cortex
Open-source libraries: OpenCV, PyTorch, Tensorflow, Blender, ONNX, GitHub, HuggingFace

SELECTIVE PUBLICATIONS

IIoT-Enabled Digital Twin or Legacy and Smart Factory Machines with LLM Integration 2025
Anuj Gautam, Manish Aryal, **Sourabh Deshpande**, Shailesh Padalkar, Ming Tang, Sam Anand, et al.
SME North American Manufacturing Research Conference (Accepted)

Deep Learning-Based Recognition of Manufacturing Components using AR for Worker Training of Assembly Tasks 2024
Sourabh Deshpande, Manish Aryal, Sam Anand
ASME Manufacturing Science and Engineering Conference Proceedings, <https://doi.org/10.1115/MSEC2024-125279>

ImVR: Immersive Design Exploration and Process Integration for Additive Manufacturing of Complex Organic Geometries 2024
Manish Aryal, **Sourabh Deshpande**, Jillian Aurisano, Sam Anand
ASME Manufacturing Science and Engineering Conference Proceedings, <https://doi.org/10.1115/MSEC2024-121253>

Smart Monitoring and Automated Real-Time Visual Inspection of a Sealant Applications 2023
Sourabh Deshpande, Aditi Roy, Joshua Johnson, Ethan Fitz, Manish Kumar, Sam Anand
SME Manufacturing Letters, <https://doi.org/10.1016/j.mfglet.2023.08.115>

IIoT Based Framework for Data Communication And Prediction Using Augmented Reality For Legacy Machine Artifacts 2023
Sourabh Deshpande, Shailesh Padalkar, Sam Anand
SME Manufacturing Letters, <https://doi.org/10.1016/j.mfglet.2023.08.058>

IIoT Based Augmented Reality for Factory Data Collection and Visualization 2021
Jonathan Rosales, **Sourabh Deshpande**, Sam Anand
Procedia Manufacturing, <https://doi.org/10.1016/j.promfg.2021.06.062>

Prediction of Selective Laser Melting Part Quality Using Hybrid Bayesian Network 2020
Nathan Hertlein, **Sourabh Deshpande**, Vysakh Venugopal, Manish Kumar, Sam Anand
Additive Manufacturing, <https://doi.org/10.1016/j.addma.2020.101089>

RELEVANT COURSEWORK

- Intelligent Systems (AI), Introduction to Industrial AI, Statistical Methods
- Math Methods for Decision Engineering, Advanced Quality Engineering, Computational Methods in Additive Manufacturing

ACADEMIC MENTORSHIP

- Manish Aryal (M.S. Thesis) – ImVR: Enabling Immersive Exploration Environment for Complex Organic Geometries, 2023
- Anuj Gautam (M.S. Thesis) – IIoT-enabled Digital Twin with LLM Feedback for Manufacturing Applications, 2024

ACCOMPLISHMENTS AND LEADERSHIP EXPERIENCE

- University of Cincinnati Office of Research Digital Futures Fellowship, 2024
- Invention Disclosure (UC), along with PI Prof. Anand and Co-PI Prof. Kumar - Object Localization MxD, 2022
- NSF student travel award for attending and recognition by SME and ASME for co-organizing NAMRC and MSEC, 2021
- Graduate teaching assistant, MECH 6071, Advanced Design for Manufacturing, 2019
- Gold and bronze medal at national level cycling tournament, School Games Federation of India, 2010
- Best All-Around and Best Character award, Tata Motors Employees Education Trust, India, 2008